

Oxytocin is a naturally occurring peptide hormone that is used clinically to prevent postpartum hemorrhage, one of the leading causes of maternal mortality globally. Because oxytocin is heat and light sensitive, it requires refrigeration at a temperature below 8 °C. However, in underdeveloped regions with hot climates (such as sub-Saharan Africa), refrigeration is often unavailable for pharmaceutical storage and transport.

In an effort to improve the availability of oxytocin in regions with limited refrigeration, researchers conducted two experiments to investigate the use of buffers and divalent alkaline-earth metal salt additives as a means to produce an aqueous oxytocin formulation that is stable when stored at temperatures up to 50 °C.

Experiment 1

To evaluate effects from buffer concentration, two sets of solutions were prepared containing 0.25 mg/mL oxytocin acetate salt in 5, 10, and 50 mM citrate buffer solution made from a mixture of citric acid (a weak acid) and sodium citrate (a salt of the conjugate base). An unbuffered aqueous solution served as a control sample. Using 1.0 M NaOH, the buffered solutions were adjusted to pH 4.5 (the pH at which oxytocin is more stable) and then stored at 5 °C (set 1) and 50 °C (set 2) for 4 weeks in a dark area. After storage, the amount of undegraded oxytocin in each sample was determined (Figure 1) using reverse-phase, high-performance liquid chromatography (RP-HPLC).

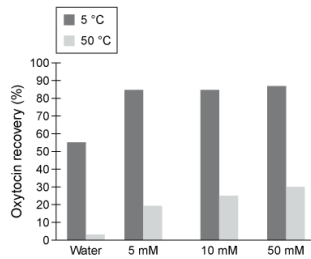


Figure 1 Oxytocin acetate recovery in a citrate buffer (pH 4.5) and water (pH 6.8)

Experiment 2

To examine the effect of divalent alkaline-earth metal cation additives, Experiment 1 was repeated, but the solutions were modified to contain 0.25 mg/mL of oxytocin acetate with 5, 10, or 50 mM of a divalent cation (from a chloride salt) dissolved in 10 mM citrate buffer. The results are shown in Figure 2.

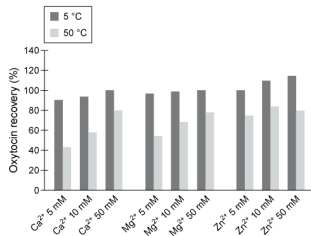


Figure 2 Oxytocin acetate in 10 mM citrate buffer with added Ca²⁺, Mg²⁺, or Zn²⁺ salts

What volume of the solutions tested would contain 0.080 g of oxytocin acetate?

- ☐ A. 0.020 mL [4%]
- ☐ B. 0.32 mL [19%]
- ☐ C. 20. mL [8%]
- ☒ D. 320 mL [67%]

Correct
67% answered correctly

Explanation:

Conversion from mass of oxytocin acetate to volume of sample

Given mass	Metric unit conversion	Sample oxytocin concentration	Volume of sample
0.080 g oxytocin	$\times \frac{1000 \text{ mg}}{1 \text{ g}}$	$\times \frac{1 \text{ mL}}{0.25 \text{ mg}}$	= 320 mL

Solution concentrations are stated in terms of the **amount of solute** per **unit volume of solution**. When the solute is a solid that has been dissolved in a liquid solvent, the solute concentration is expressed in terms of **moles or mass of solute** per **unit volume of solution**. In medicine, the solute is often an active ingredient such as a drug. To determine the volume of a solution required to deliver a desired amount of an active ingredient (solute), the desired amount of solute must be converted to matching units of mass (or moles) and then divided by the solution concentration.

In the experiments, the concentration of oxytocin acetate in each of the samples is 0.25 mg/mL. The question asks what volume of solution would contain 0.080 g of oxytocin acetate. Because the concentration is given in terms of milligrams per milliliter and the specified mass of oxytocin acetate is given in grams, the first step is to convert the specified mass to milligrams:

$$0.080 \text{ g oxytocin} \times \frac{1000 \text{ mg}}{1 \text{ g}} = 80. \text{ mg oxytocin}$$

Next, dividing by the solution concentration gives the required solution volume:

$$80. \text{ mg oxytocin} \times \frac{1 \text{ mL solution}}{0.25 \text{ mg oxytocin}} = 320 \text{ mL solution}$$

The answer choices are in terms of milliliters, so no other calculations are needed to determine volume.

(Choice A) A value of 0.020 (with incorrect units) is obtained by failing to convert 0.080 g to milligrams and then multiplying rather than dividing this mass by the solution concentration.

(Choice B) A value of 0.32 (with incorrect units) results from not converting 0.080 g to milligrams. Mismatched units of mass do not cancel during the calculation.

(Choice C) A result of 20. (with incorrect units) is found if the mass (in milligrams) is multiplied rather than divided by the solution concentration.

Educational objective:

Solution concentrations are given in terms of the amount of solute per unit volume of solution. Solute amounts are typically given in terms of moles or mass, whereas solvent amounts are given in terms of volume. The volume of a solution containing a certain amount of solute is found by converting the solute amount to matching units of mass (or moles) and then dividing by the solution concentration.

Time spent: 0 sec

Subject:
General Chemistry

QID: 402639

System:
Solutions and Electrochemistry

Oxytocin is a naturally occurring peptide hormone that is used clinically to prevent postpartum hemorrhage, one of the leading causes of maternal mortality globally. Because oxytocin is heat and light sensitive, it requires refrigeration at a temperature below 8 °C. However, in underdeveloped regions with hot climates (such as sub-Saharan Africa), refrigeration is often unavailable for pharmaceutical storage and transport.

In an effort to improve the availability of oxytocin in regions with limited refrigeration, researchers conducted two experiments to investigate the use of buffers and divalent alkaline-earth metal salt additives as a means to produce an aqueous oxytocin formulation that is stable when stored at temperatures up to 50 °C.

Experiment 1

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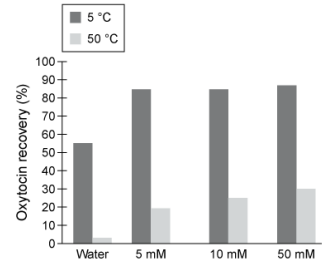


Figure 1 Oxytocin acetate recovery in a citrate buffer (pH 4.5) and water (pH 6.8)

Experiment 2

To examine the effect of divalent alkaline-earth metal cation additives, Experiment 1 was repeated, but the solutions were modified to contain 0.25 mg/mL of oxytocin acetate with 5, 10, or 50 mM of a divalent cation (from a chloride salt) dissolved in 10 mM citrate buffer. The results are shown in Figure 2.

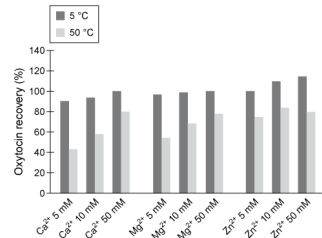


Figure 2 Oxytocin acetate in 10 mM citrate buffer with added Ca²⁺, Mg²⁺, or Zn²⁺ salts

Suppose the 10 mM MgCl₂ solution used in Experiment 2 is replaced with a 5 mM NaCl solution. Compared to the Cl⁻ concentration in the 10 mM MgCl₂ solution, the concentration of Cl⁻ in the 5 mM NaCl solution would be:

- ☐ A. greater. [3%]
- ☐ B. the same. [10%]
- ☐ C. 2 times less. [19%]
- ☒ D. 4 times less. [66%]

Correct

67% answered correctly

Explanation:

NaCl solution

NaCl solution concentration

5 mmol NaCl
1 L solution

Mole ratio of Cl⁻ ion

1 mmol Cl⁻
1 mmol NaCl

=

Concentration of Cl⁻ in solution

5 mM Cl⁻

MgCl₂ solution

MgCl₂ solution concentration

10 mmol MgCl₂
1 L solution

Mole ratio of Cl⁻ ion

2 mmol Cl⁻
1 mmol MgCl₂

=

Concentration of Cl⁻ in solution

20 mM Cl⁻

The 5 mM NaCl solution has a Cl⁻ ion concentration that is 4 times lower than the 10 mM MgCl₂ solution.

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The **millimolar concentration** (mM) of a solution (ie, the **Molarity (M)** scaled down by a factor of 1,000) is defined as millimoles of solute per liter of solution:

$$\text{mM} = \frac{\text{millimoles of solute}}{\text{liter of solution}} = 1 \times 10^{-3} \text{ M}$$

Because a millimolar concentration is defined in terms of moles, it can be used with **stoichiometric mole ratios** from **balanced chemical reactions** or compound formulas to compare the amount of one solute to the amount of another solute in a given volume of a solution. The general chemical reaction for the dissolution of strong electrolytes in water is:

$$X_mY_n \rightarrow mX^{n+} + nY^{m-}$$

where *m* and *n* are the **stoichiometric coefficients** resulting from the charges of each ion. Accordingly, the equation for the dissolution of NaCl shows a 1:1 mole ratio (or 1:1 millimole ratio) of Cl⁻ to NaCl:

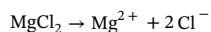
$$\text{NaCl} \rightarrow \text{Na}^{+} + \text{Cl}^{-}$$

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Multiplying the concentration of the NaCl solution by this mole ratio yields the concentration of Cl⁻ ions present in the solution:

$$\frac{5 \text{ mmol NaCl}}{1 \text{ L solution}} \times \frac{1 \text{ mmol Cl}^-}{1 \text{ mmol NaCl}} = \frac{5 \text{ mmol Cl}^-}{1 \text{ L solution}} = 5 \text{ mM Cl}^-$$

In contrast, the equation for the dissolution of MgCl₂ shows a 2:1 mole ratio (or 2:1 millimole ratio) of Cl⁻ to MgCl₂:



Consequently, the concentration of Cl⁻ ions present in the 10 mM MgCl₂ solution is:

$$\frac{10 \text{ mmol MgCl}_2}{1 \text{ L solution}} \times \frac{2 \text{ mmol Cl}^-}{1 \text{ mmol MgCl}_2} = \frac{20 \text{ mmol Cl}^-}{1 \text{ L solution}} = 20 \text{ mM Cl}^-$$

Dividing the concentration of Cl⁻ in the NaCl solution by the concentration of Cl⁻ in the MgCl₂ solution gives:

$$\frac{5 \text{ mM}}{20 \text{ mM}} = \frac{1}{4}$$

Therefore, the concentration of Cl⁻ in the 5 mM NaCl solution is 4 times less than the concentration of Cl⁻ in the 10 mM MgCl₂ solution.

(Choices A and B) The concentration of Cl⁻ ions is *lower* in the NaCl solution because the concentration of NaCl is lower than MgCl₂ and because the mole ratio of Cl⁻:NaCl is 1:1, whereas the mole ratio of Cl⁻:MgCl₂ is 2:1.

(Choice C) A concentration that is two times less is obtained when only the compound concentrations, but not the mole ratios of the compounds to Cl⁻, are considered.

Educational objective:

The millimolar concentration of a solution is defined as the number of millimoles of solute per liter of solution. Stoichiometric mole ratios and the millimolar concentrations can be used to assess the relative molar amounts of different species in an aliquot of solution.

Time spent: 0 sec
Subject:
General Chemistry

QID: 402640
System:
Solutions and Electrochemistry

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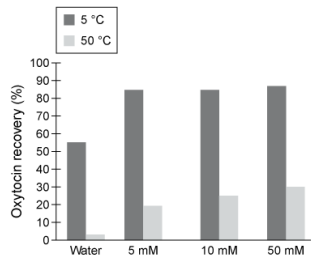


Figure 1 Oxytocin acetate recovery in a citrate buffer (pH 4.5) and water (pH 6.8)

Experiment 2

To examine the effect of divalent alkaline-earth metal cation additives, Experiment 1 was repeated, but the solutions were modified to contain 0.25 mg/mL of oxytocin acetate with 5, 10, or 50 mM of a divalent cation (from a chloride salt) dissolved in 10 mM citrate buffer. The results are shown in Figure 2.

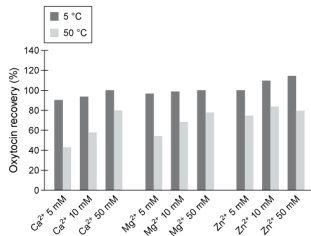


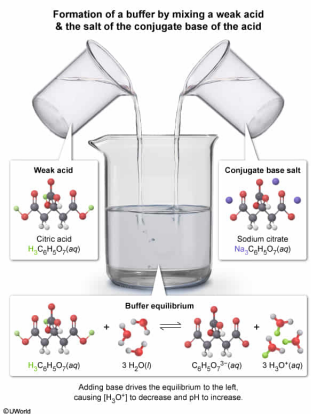
Figure 2 Oxytocin acetate in 10 mM citrate buffer with added Ca²⁺, Mg²⁺, or Zn²⁺ salts

Which of the following experimental changes to the initially prepared oxytocin acetate solutions will decrease the percent of oxytocin acetate recovered in Experiment 1 after the 4-week storage period?

- ☒ A. Adding more sodium citrate after the NaOH has been added [53%]
- ☐ B. Decreasing the initial concentration of oxytocin acetate [33%]
- ☐ C. Storing the solutions at 0 °C rather than 5 °C [5%]
- ☐ D. Increasing the citrate buffer concentration [7%]

Correct
53% answered correctly

Explanation:



A **buffered solution** consists of a mixture of a **weak acid (HA)** and a salt of its **conjugate base (A⁻)** or a mixture of a **weak base** and its conjugate acid. Buffered solutions **resist changes in pH** by **neutralizing** H⁺ or OH⁻ ions when small amounts of a **strong acid** or **strong base** are added. The equilibrium expression for the **dissociation** of a weak acid is:

$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

where K_a is the acid dissociation constant and $[H^+]$, $[A^-]$, and $[HA]$ are the molar concentrations of the chemical species at equilibrium. Rearranging this expression to solve for $[H^+]$ and taking the negative logarithm of both sides of the equation yields the relationship between pH, $[HA]$, and $[A^-]$:

$$\begin{aligned}[H^+] &= K_a \frac{[HA]}{[A^-]} \\ -\log[H^+] &= -\log\left(K_a \frac{[HA]}{[A^-]}\right) \\ \text{pH} &= \text{p}K_a + \log\left(\frac{[A^-]}{[HA]}\right)\end{aligned}$$

Accordingly, the pH of a buffered solution depends on the concentration ratio of acid to conjugate base. Adding more acid will decrease the pH and adding more base will increase the pH.

In Experiment 1, the citrate buffer is made of citric acid $[HA]$ and sodium citrate $[A^-]$. Adding more sodium citrate to the initially prepared solutions (pH of 4.5) will increase the pH of the solution (pH > 4.5). The passage states that oxytocin is more stable at a pH of 4.5, and Figure 1 shows that at the higher pH of 6.8 (water), less oxytocin was recovered. Therefore, adding more sodium citrate will decrease the percent recovery of oxytocin acetate.

(Choice B) The experiment measures the *percent recovery* of oxytocin. Decreasing the initial amount of oxytocin acetate dissolved in solution will lower the total amount present in solution but will not change the *percentage* recovered.

(Choice C) Storing the solutions at 0 °C rather than 5 °C during the experiment will not decrease the percent recovery because the results in Experiments 1 and 2 show that oxytocin acetate is *more stable* at cooler temperatures.

(Choice D) Increasing the buffer concentration will *increase* buffer capacity to resist changes in pH and assist in stabilizing the oxytocin acetate. Based on the results in Experiment 1, this will not decrease oxytocin acetate recovery.

Educational objective:

A buffered solution consists of a weak acid and a salt of its conjugate base or a weak base and its conjugate acid. Buffers resist changes in pH by neutralizing added H^+ and OH^- ions. The pH of a buffered solution depends on the ratio of weak acid and its conjugate base.

Time spent: 0 sec
Subject:
General Chemistry

QID: 402641
System:
Solutions and Electrochemistry

Oxytocin is a naturally occurring peptide hormone that is used clinically to prevent postpartum hemorrhage, one of the leading causes of maternal mortality globally. Because oxytocin is heat and light sensitive, it requires refrigeration at a temperature below 8 °C. However, in underdeveloped regions with hot climates (such as sub-Saharan Africa), refrigeration is often unavailable for pharmaceutical storage and transport.

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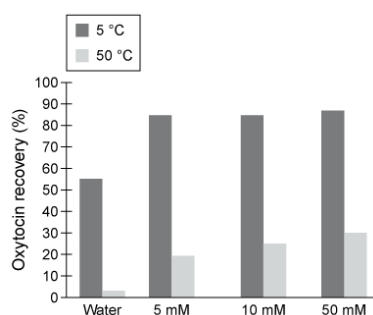


Figure 1 Oxytocin acetate recovery in a citrate buffer (pH 4.5) and water (pH 6.8)

Experiment 2

To examine the effect of divalent alkaline-earth metal cation additives, Experiment 1 was repeated, but the solutions were modified to contain 0.25 mg/mL of oxytocin acetate with 5, 10, or 50 mM of a divalent cation (from a chloride salt) dissolved in 10 mM citrate buffer. The results are shown in Figure 2.

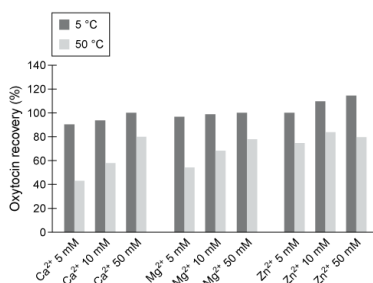


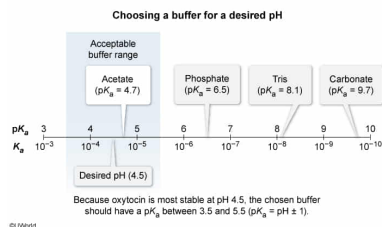
Figure 2 Oxytocin acetate in 10 mM citrate buffer with added Ca²⁺, Mg²⁺, or Zn²⁺ salts

Based on the passage, which of the following buffers could be used instead of the citrate buffer in Experiments 1 and 2?

- ✓ ☒ A. Acetate (pK_a 4.7) [82%]
☐ B. Phosphate (pK_a 6.5) [9%]
☐ C. Tris (pK_a 8.1) [2%]
☐ D. Carbonate (pK_a 9.7) [5%]

Correct
81% answered correctly

Explanation:



A **buffer solution** consists of a mixture of a **weak acid** and a salt of its **conjugate base** or a mixture of a **weak base** and its **conjugate acid**. Each component of the buffer acts to **neutralize** any added acid or base by either protonating OH^- ions, or by bonding H^+ ions. Because of these interactions, buffered systems are resistant to large changes in pH.

The **acid dissociation constant** K_a of a weak acid indicates the **relative strength** of the acid. For convenience, this value is often expressed as the pK_a , which is defined as the negative logarithm of the K_a . When there are **equal concentrations** of the weak acid (or weak base) and its conjugate in a buffer solution, **$\text{pH} = pK_a$** .

A buffer solution has a limited pH range, known as the **buffering range**, in which it is most effective. The **buffering range** typically lies within one pH unit of the pK_a (**$\text{pH range} = pK_a \pm 1$**). Accordingly, the buffer solution chosen for a particular experiment should have a

pK_a near the desired pH for the experiment.

The passage states that oxytocin is most stable at pH 4.5. This means the best buffer for oxytocin will have a pK_a between 3.5 and 5.5. Of the given choices, acetate has a pK_a (4.7) that is closest to the desired pH.

(Choices B, C, and D) Each of these buffers has a pK_a outside the desired range ($3.5 < pK_a < 5.5$).

Educational objective:

A buffered solution consists of a mixture of a weak acid and a salt of its conjugate base or a mixture of a weak base and its conjugate acid. When the concentrations of the weak acid (or weak base) and its conjugate are equal, $\text{pH} = pK_a$. The best buffer for maintaining a given pH should have a pK_a that is within one unit of the desired pH.

Time spent: 0 sec

Subject:
General Chemistry

QID: 402642

System:
Solutions and Electrochemistry

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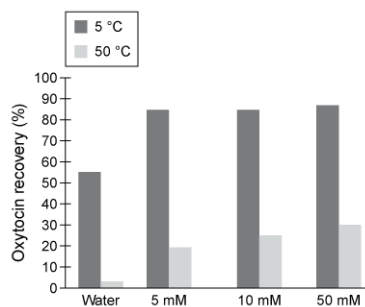


Figure 1 Oxytocin acetate recovery in a citrate buffer (pH 4.5) and water (pH 6.8)

Experiment 2

To examine the effect of divalent alkaline-earth metal cation additives, Experiment 1 was repeated, but the solutions were modified to contain 0.25 mg/mL of oxytocin acetate with 5, 10, or 50 mM of a divalent cation (from a chloride salt) dissolved in 10 mM citrate buffer. The results are shown in Figure 2.

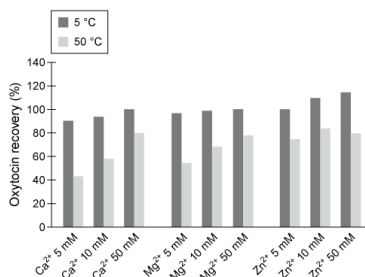


Figure 2 Oxytocin acetate in 10 mM citrate buffer with added Ca²⁺, Mg²⁺, or Zn²⁺ salts

If 16 mL of the initial oxytocin acetate solution containing the 50 mM Zn²⁺ additive was analyzed, how many moles of oxytocin acetate (MW = 1067 g/mol) were recovered from the initial sample after 4 weeks at 50 °C?

- ☐ A. 4.8×10^{-5} mol [6%]
- ☐ B. 4.6×10^{-6} mol [23%]
- ☒ C. 3.0×10^{-6} mol [60%]
- ☐ D. 1.1×10^{-6} mol [9%]

Correct
58% answered correctly

Explanation:

Determine the number of moles of oxytocin acetate recovered

Initial volume analyzed	Concentration oxytocin acetate	Unit conversion	Molar mass oxytocin acetate	80% oxytocin acetate recovery	Moles recovered
16 mL	$\frac{0.25 \text{ mg}}{1 \text{ mL}}$	$\frac{1 \text{ g}}{1000 \text{ mg}}$	$\frac{1 \text{ mol}}{1067 \text{ g}}$	$\frac{80 \text{ mol recovered}}{100 \text{ mol original}}$	$3.0 \times 10^{-6} \text{ mol oxytocin acetate}$

CUSTOM

Solution concentration can sometimes be expressed in terms of **mass per unit volume**. When concentration is stated in this way, the solution concentration can be used as a conversion factor to find the **volume** of solution that contains a given amount of **mass**. If the number of moles of a substance is known instead of the mass, the moles can be **converted** into the corresponding amount of mass using the **molar mass**.

The passage states that each sample solution containing the Zn^{2+} additive (used in Experiment 2) contains 0.25 mg/mL of oxytocin acetate. Using the given solution concentration as a conversion factor, the initial mass oxytocin acetate in the analyzed solution is

$$16 \text{ mL solution} \times \frac{0.25 \text{ mg}}{1 \text{ mL solution}} \times \frac{1 \text{ g}}{1000 \text{ mg}} = 4.0 \times 10^{-3} \text{ g oxytocin acetate}$$

Converting grams of oxytocin acetate to the corresponding number of moles using its molar mass gives:

$$4.0 \times 10^{-3} \text{ g} \times \frac{1 \text{ mol}}{1067 \text{ g}} = 3.7 \times 10^{-6} \text{ mol oxytocin acetate}$$

According to Figure 2, approximately 80% of the initial amount of oxytocin acetate was recovered from the sample containing the 50 mM Zn^{2+} additive after being stored for 4 weeks at 50 °C. As such, if 3.7×10^{-6} moles of oxytocin acetate were initially present in the sample, the **amount recovered** after being stored is:

$$(3.7 \times 10^{-6} \text{ mol}) \times \left(\frac{80}{100}\right) = (3.7 \times 10^{-6} \text{ mol}) \times (0.80) = 3.0 \times 10^{-6} \text{ mol oxytocin acetate}$$

(Choice A) A value of 4.8×10^{-5} mol (with incorrect units) is obtained by dividing, rather than multiplying, the volume of analyzed solution by the solution concentration of 0.25 mg/mL.

(Choice B) A value of 4.6×10^{-6} mol is found if the moles of oxytocin acetate initially in the sample is divided, rather than multiplied, by 0.80 when calculating the moles recovered after being stored.

(Choice D) A value of 1.1×10^{-6} mol is calculated if 3.7×10^{-6} mol is multiplied by 0.30 (from Figure 1 for a sample without any Zn^{2+} additive) rather than by 0.80 (from Figure 2 for the solution containing the 50 mM Zn^{2+} additive).

Educational objective:

Solution concentration can be expressed in terms of mass per unit volume, which can be used as a conversion factor to find the volume of solution that contains a given amount of mass. Conversions between the moles and mass of a substance can be made using the molar mass.

Time spent: 0 sec
Subject:
General Chemistry

QID: 402643
System:
Solutions and Electrochemistry

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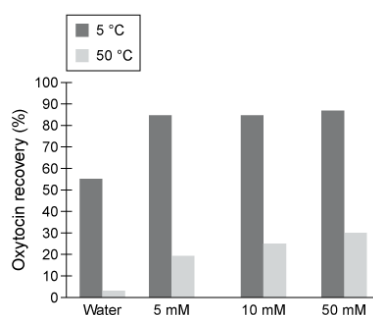


Figure 1 Oxytocin acetate recovery in a citrate buffer (pH 4.5) and water (pH 6.8)

Experiment 2

To examine the effect of divalent alkaline-earth metal cation additives, Experiment 1 was repeated, but the solutions were modified to contain 0.25 mg/mL of oxytocin acetate with 5, 10, or 50 mM of a divalent cation (from a chloride salt) dissolved in 10 mM citrate buffer. The results are shown in Figure 2.

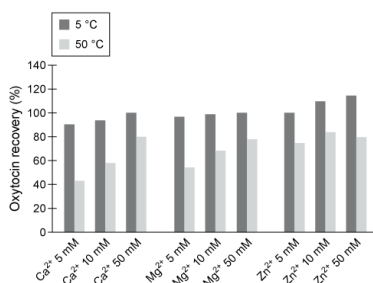
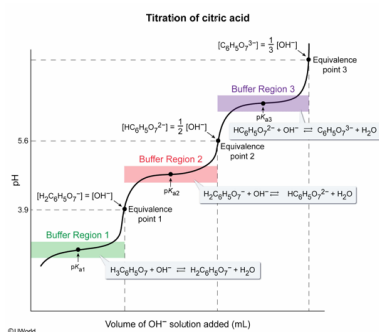


Figure 2 Oxytocin acetate in 10 mM citrate buffer with added Ca²⁺, Mg²⁺, or Zn²⁺ salts

Suppose that citric acid (H₃C₆H₅O₇) is titrated with 0.1 M NaOH to form a citrate buffer solution with a pH of 4.5. What is the pH at the first equivalence point? (Note: pK_{a1} = 3.13, pK_{a2} = 4.76, pK_{a3} = 6.40)

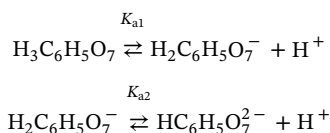
- ☐ A. Less than 3.13 [6%]
- ✓ ☒ B. Between 3.13 and 4.76 [69%]
- ☐ C. Equal to 3.13 [20%]
- ☐ D. Greater than 4.76 [3%]

Explanation:



During the titration of a weak **polyprotic acid** with a **strong base**, the first stoichiometric **equivalence point** occurs when the moles of added base are equal to the moles of the initial acid species. Prior to the equivalence point, a **buffering region** forms in which both the **weak acid** and its **conjugate base** are present. The **pH** of the solution at any point in the buffering region is determined by the pK_a of the acid and the **ratio** of the acid and conjugate base species concentrations. At the midpoint of the buffer region (ie, the **half-equivalence point**), the concentrations of the weak acid and its conjugate base are equal and **pH = pK_a** .

In this question, at the first equivalence point, the initial triprotic acid species ($\text{H}_3\text{C}_6\text{H}_5\text{O}_7$) has been fully consumed and converted into its conjugate base ($\text{H}_2\text{C}_6\text{H}_5\text{O}_7^-$). For a polyprotic acid, the first conjugate base species formed is **amphoteric** and can act as either an acid (by donating an additional proton) or as a base (by accepting a proton and regenerating the original acid). Consequently, the pH at the first equivalence point must be between the pK_a of the first acidic hydrogen (pK_{a1}) and the second acidic hydrogen (pK_{a2}) because the two equilibrium reactions occurring in solution are:



Therefore, the pH of the $\text{H}_3\text{C}_6\text{H}_5\text{O}_7$ solution at the first equivalence point must be greater than 3.13 (pK_{a1}) but less than 4.76 (pK_{a2}).

Note that the exact pH can be found by averaging the pK_a values:

$$\frac{3.13 + 4.76}{2} = 3.9$$

(Choice A) The pH will be greater, not less, than pK_{a1} (3.13) because at the equivalence point, all of the $\text{H}_3\text{C}_6\text{H}_5\text{O}_7$ species have been deprotonated once by OH^- .

(Choice C) The pH at the half-equivalence point for a monoprotic acid is equal to pK_{a1} (3.13). However, with polyprotic acids, the pH at the first equivalence point is determined by both pK_{a1} and pK_{a2} .

(Choice D) The pH at the first equivalence point will be *less than* pK_{a2} (4.76) because at the first equivalence point, additional acidic protons are present within the structure and can dissociate into solution.

Educational objective:

In a titration of a weak polyprotic acid and a strong base, the pH at a stoichiometric equivalence point is determined by the pK_a values of the acid and the conjugate base species present in the equilibrium reaction.

Time spent: 0 sec
Subject:
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